Exercise 7.1. *Complete the proof of the theorem using the C matrix proposed above and find the constant a along the way. Hint: You may need Eq. (7.2).*

$$M(g_1g_2) := \rho_1(g_1)M(g_2) + M(g_1)\rho_2(g_2)$$

$M(g) + C\rho_2(g) = \rho_1(g)C \quad \forall g \in G \qquad \rho(g_1)\rho(g_2) = \rho(g_1g_2)$

$C = a\sum_{h\in G} M(h^{-1})\rho_2(h)$

$M(g) + a\sum_{h\in G} M(h^{-1})\rho_2(h)\rho_2(g) = a\sum_{h\in G}\rho_1(g)M(h^{-1})\rho_2(h)$

$M(g) + a\sum_{h\in G}\left[M(h^{-1})\rho_2(hg) - \rho_1(g)M(h^{-1})\rho_2(h)\right] = 0$

since $M(gh^{-1}) = \rho_1(g)M(h^{-1}) + M(g)\rho_2(h^{-1})$

$M(g) + a\sum_{h\in G}\left[\underline{M(h^{-1})\rho_2(hg)} - \left[\underline{M(gh^{-1})} - M(g)\rho_2(h^{-1})\right]\underline{\rho_2(h)}\right] = 0$

$\sum_{h\in G} M(h^{-1})\rho_2(hg) = \sum_{h\in G} M(gh^{-1})\rho_2(h)$ according to rearrangement

$M(g) + a\sum_{h\in G} M(g) = 0 \qquad a = -\frac{1}{|G|}$

Exercise 7.3. *For any group element $g \in G$, the set $Z_g = \{h \in G | hg = gh\}$ of all the group elements that commute with g form a subgroup of G, called a* ***centralizer subgroup****. Prove that Z_g is indeed a subgroup of G. Prove that if $g, g' \in [g]$, then $Z_g \cong Z_{g'}$. Hence, centralizer subgroups can be indexed by the conjugacy classes C^A of G and denoted by $Z_A := Z_{g^A}$, where g^A is a chosen representative of the conjugacy class C^A.*

(1) ① $e \in Z_g$ ② if $h \in Z_g$, $hg = gh$, 则 $g = h^{-1}gh$ $\quad gh^{-1} = h^{-1}g$, $h^{-1} \in Z_g$

③ if $h_1, h_2 \in Z_g$. $h_1h_2g = h_1gh_2 = gh_1h_2$, $h_1h_2 \in Z_g$. 封闭

(2) $\exists f \in G$, $g' = fgf^{-1}$ 对 $\forall h \in Z_g$, $hg = gh$

$\exists h' \in Z_{g'}$, $h'g' = g'h'$ $\quad h'fgf^{-1} = fgf^{-1}h'$ $\quad f^{-1}h'fg = gf^{-1}h'f$

$\Rightarrow h = f^{-1}h'f$

反之亦然. 因此 $Z_g \mapsto Z_{g'}$ 是双射. $Z_g \cong Z_{g'}$

Exercise 7.5. *In S_n, a k-cycle $(a_1a_2\cdots a_k)$ is defined by the permutation $a_1 \to a_2 \to \ldots \to a_k \to a_1$, where a_1 through a_k are certain k objects out of the total n objects. You have seen examples of cycles in S_3 above. Now answer the following questions.*

1. *How many k-cycles are there in S_n?* $C_n^k\, k!$
2. *What is the inverse of a k-cycle $(a_1a_2\cdots a_k)$?* $\begin{pmatrix} a_1 & a_2 & \cdots & a_k \\ a_2 & a_3 & \cdots & a_1 \end{pmatrix}\begin{pmatrix} a_1 & a_2 & \cdots & a_k \\ a_k & a_1 & \cdots & a_{k-1} \end{pmatrix} = 1 \quad (a_1a_2\cdots a_k)^{-1} = (a_ka_{k-1}\cdots a_1)$
3. *Prove that $(a_1a_2\cdots a_k)^k = e$, the identity element.* $\begin{pmatrix} a_1 & a_2 & \cdots & a_k \\ a_2 & a_3 & \cdots & a_1 \end{pmatrix}\begin{pmatrix} a_1 & a_2 & \cdots & a_k \\ a_2 & a_3 & \cdots & a_1 \end{pmatrix}\cdots\begin{pmatrix} a_1 & a_2 & \cdots & a_k \\ a_2 & a_3 & \cdots & a_1 \end{pmatrix}$，追踪$a_1$的动向：
$a_1 \xrightarrow{1} a_2 \xrightarrow{2} a_3 \xrightarrow{3} a_4 \cdots \xrightarrow{k} a_1$
其余亦然.
4. *Two cycles $(a_1a_2\cdots a_m)$ and $(a_1'a_2'\cdots a_n')$ are disjoint if $\{a_1, a_2, \cdots, a_m\} \cap \{a_1', a_2', \cdots, a_n'\} = \emptyset$. Prove that two disjoint cycles commute with each other.*
5. *A k-cycle is called a **transposition** if $k = 2$. Show that any transposition can be written as $(mn) = (1m)(1n)(1m)$. Then, show that any cycle can be written as the product of some transpositions. Transpositions $(1m)$'s are therefore the generators of the permutation group.*
6. *Show that any permutation in S_n can be uniquely written as a product of some disjoint cycles, in which there are m_j j-cycles with $j = 1, 2, \cdots, n$. This is called the cycle structure of a permutation, which is characterized by the set of numbers $\{m_j \geq 0 | j = 1, \ldots, n\}$. What is the relation among the m_j's in a cycle structure?*
7. *What kinds of cycle structures can S_4 have? (What is the relation between cycle structures and partitions of integers?)* 相等
$(****)\ (***)(*)\ (**)(**)\ (**)(*)(*)$
$(*)(*)(*)(*)$ $\quad n = p(4) = 5$
8. *In S_n, how many elements have a given cycle structure $\{m_j\}$?* m_j 个 j-cycle $\quad n_{\{m_j\}} = \dfrac{n!}{\prod_{j=1}^{n}(m_j!\, j^{m_j})}$
9. *Prove that permutations in the same conjugacy class have the same cycle structure.*

4. $(a_1a_2\cdots a_m)(a_1'a_2'\cdots a_n') = \begin{pmatrix} a_1 & a_2 & \cdots & a_m & \cdots & a_1' & a_2' & \cdots & a_n' & \cdots \\ a_2 & a_3 & \cdots & a_1 & \cdots & a_2' & a_3' & \cdots & a_1' & \cdots \end{pmatrix} = (a_1'a_2'\cdots a_n')(a_1a_2\cdots a_m)$

5. $(1m)(1n)(1m) = \begin{pmatrix} 1 & \cdots & m & \cdots & n & \cdots \\ m & \cdots & 1 & \cdots & n & \cdots \end{pmatrix}\begin{pmatrix} 1 & \cdots & m & \cdots & n & \cdots \\ n & \cdots & m & \cdots & 1 & \cdots \end{pmatrix}\begin{pmatrix} 1 & \cdots & m & \cdots & n & \cdots \\ m & \cdots & 1 & \cdots & n & \cdots \end{pmatrix}$

$= \begin{pmatrix} 1 & \cdots & m & \cdots & n & \cdots \\ 1 & \cdots & n & \cdots & m & \cdots \end{pmatrix} = (mn)$

any cycle $(a_1a_2\cdots a_m) = \begin{pmatrix} a_1 & a_2 & \cdots & a_m \\ a_2 & a_3 & \cdots & a_1 \end{pmatrix} = (m-1, m)\cdots(2m)(1m)$
$= (12)(23)\cdots(m-1, m)$

6. any permutation $\begin{pmatrix} 1 & 2 & \cdots & n \\ \downarrow & & \downarrow & \\ a_1 & a_2 & \cdots & a_n \end{pmatrix}$

can be interpreted as $1 \to a_1$, $a_1 \to a_1' \cdots$, until it returns to 1

therefore $\begin{pmatrix} 1 & 2 & \cdots & n \\ a_1 & a_2 & \cdots & a_n \end{pmatrix} = (**)(\quad)\cdots$

9. given to permutations $P = \begin{pmatrix} 1 & 2 & \cdots & n \\ a_1 & a_2 & \cdots & a_n \end{pmatrix} \quad Q = \begin{pmatrix} 1 & 2 & \cdots & n \\ b_1 & b_2 & \cdots & b_n \end{pmatrix}$

$P' = QPQ^{-1} = \begin{pmatrix} 1 & 2 & \cdots & n \\ b_1 & b_2 & \cdots & b_n \end{pmatrix}\begin{pmatrix} 1 & 2 & \cdots & n \\ a_1 & a_2 & \cdots & a_n \end{pmatrix}\begin{pmatrix} b_1 & b_2 & \cdots & b_n \\ 1 & 2 & \cdots & n \end{pmatrix}$

$= \begin{pmatrix} a_1 & a_2 & \cdots & a_n \\ c_1 & c_2 & \cdots & c_n \end{pmatrix}\begin{pmatrix} 1 & 2 & \cdots & n \\ a_1 & a_2 & \cdots & a_n \end{pmatrix}\begin{pmatrix} b_1 & b_2 & \cdots & b_n \\ 1 & 2 & \cdots & n \end{pmatrix}$

$= \begin{pmatrix} b_1 & b_2 & \cdots & b_n \\ c_1 & c_2 & \cdots & c_n \end{pmatrix}$

因此共轭变换是对P的上、下行同时施行Q变换得到的结果，

P和P'轮换结构未变，只是具体数字发生置换

Exercise 7.7. *Find the similarity transformation that can transform the matrices in Eq. (7.10) and those in Eq. (7.13)*

$$\rho^R((12)) = \begin{pmatrix} 0&1&0&0&0&0\\ 1&0&0&0&0&0\\ 0&0&0&0&1&0\\ 0&0&0&0&0&1\\ 0&0&1&0&0&0\\ 0&0&0&1&0&0 \end{pmatrix}, \quad \rho^R((123)) = \begin{pmatrix} 0&0&0&0&0&1\\ 0&0&1&0&0&0\\ 0&0&0&1&0&0\\ 0&1&0&0&0&0\\ 1&0&0&0&0&0\\ 0&0&0&0&1&0 \end{pmatrix} \tag{7.10}$$

A

$$\rho^{3\otimes 2}((12)) = \begin{pmatrix} 0&0&-1&0&0&0\\ 0&0&0&1&0&0\\ -1&0&0&0&0&0\\ 0&1&0&0&0&0\\ 0&0&0&0&-1&0\\ 0&0&0&0&0&1 \end{pmatrix}, \quad \rho^{3\otimes 2}((123)) = \begin{pmatrix} 0&0&0&0&-\frac{1}{2}&-\frac{\sqrt{3}}{2}\\ 0&0&0&0&\frac{\sqrt{3}}{2}&-\frac{1}{2}\\ -\frac{1}{2}&-\frac{\sqrt{3}}{2}&0&0&0&0\\ \frac{\sqrt{3}}{2}&-\frac{1}{2}&0&0&0&0\\ 0&0&-\frac{1}{2}&-\frac{\sqrt{3}}{2}&0&0\\ 0&0&\frac{\sqrt{3}}{2}&-\frac{1}{2}&0&0 \end{pmatrix}, \tag{7.13}$$

B

$A \sim B \qquad PAP^{-1} = \Lambda \qquad QBQ^{-1} = \Lambda$

$B = Q^{-1}P\Lambda P^{-1}Q$

Exercise 7.9. *For any unitary representation (ρ, v) of a finite group G, using projectors to prove that if ρ is reducible, it is fully reducible.*

设ρ为n维表示 $\quad |V_0\rangle = \frac{1}{\sqrt{n}}(|1\rangle + |2\rangle + \cdots + |n\rangle)$

$P_1 = |V_0\rangle\langle V_0|$

$P_2 = \mathbb{1}_n - P_1$ 有$(n-1)$本征矢 $|V_1\rangle \cdots |V_{n-1}\rangle$

$|V_0\rangle' = \frac{1}{\sqrt{n-1}}(|V_1\rangle + \cdots + |V_{n-1}\rangle)$

$P_1' = |V_0\rangle'\langle V_0|'$

最后可找出n个 projectors，即 fully reducible

Exercise 7.11. *Show that $\rho\rho^\dagger = \mathbb{1}_{\dim V}$ too.*

$D = UHU^{-1} \qquad D^{-1} = UH^{-1}U^{-1} \qquad H = \sum_{g\in G} \tilde{\rho}^\dagger(g)\tilde{\rho}(g)$

$H^{-1} = \sum_{g\in G} \tilde{\rho}^{-1}(g)\tilde{\rho}^{-\dagger}(g)$

$\tilde{\rho}(g')H^{-1}\tilde{\rho}^\dagger(g')$

$= \sum_{g\in G} \tilde{\rho}(g'g^{-1})\tilde{\rho}^\dagger(g'g^{-1}) = H^{-1}$

$$\rho\rho^\dagger = \sqrt{D}U\tilde{\rho}\underbrace{U^{-1}\sqrt{D}^{-1}\sqrt{D}^{-1}U}_{H^{-1}}\tilde{\rho}^\dagger U^{-1}\sqrt{D}$$

$$= \sqrt{D}U\underbrace{\tilde{\rho}H^{-1}\tilde{\rho}^\dagger}_{H^{-1}}U^{-1}\sqrt{D}$$

$$= \sqrt{D}\underbrace{UH^{-1}U^{-1}}_{D^{-1}}\sqrt{D} = \mathbb{1}_{\dim V}$$

Exercise 7.13. *Apply the reducibility checkers (7.20) and (7.22) to verify that the ρ^2 of S_3 is irreducible.*

$\chi^2(e)=2 \qquad \chi^2([(12)])=0 \qquad \chi^2([(123)])=-1$

$$\sum_{A=1}^{n_C} n_A \chi_A^* \chi_A = \sum_{\mu=1}^{r}\sum_{\nu=1}^{r} n_\mu n_\nu \sum_{A=1}^{n_C} n_A \chi_A^{\mu *} \chi_A^{\nu} = \sum_{\mu=1}^{r}\sum_{\nu=1}^{r} n_\mu n_\nu \delta^{\mu\nu}|G| = |G| \sum_{\mu=1}^{r} n_\mu^2. \quad (7.20)$$

$$\frac{1}{|G|}\sum_{A=1}^{n_C} n_A \chi_A^{\mu *} \chi_A = \frac{1}{|G|}\sum_{A=1}^{n_C} n_A \chi_A^{\mu *} \sum_{\nu=1}^{r} n_\nu \chi_A^{\nu} = \sum_{\nu=1}^{r} n_\nu \delta^{\mu\nu} = n_\mu. \tag{7.22}$$

$$\sum_{A=1}^{n_C} n_A \chi_A^* \chi_A = 1\times 2^2 + 3\times 0^2 + 2\times(-1)^2 = 6 = 6\times 1^2$$

$$\frac{1}{|G|}\sum_{A=1}^{n_C} n_A \chi_A^{\mu *} \chi_A = 1 = n_\mu$$

Exercise 7.15. *Consider an N-periodic function $x(n)$, defined by*

$$x(n) = \begin{cases} 1, & 0 \le n \le k-1, \\ 0, & k \le n \le N-1. \end{cases}$$

What is the discrete Fourier transform of $x(n)$? Use Parseval's theorem to compute the sum

$$\sum_{\mu=1}^{N-1} \frac{1-\cos\frac{2\pi\mu k}{N}}{1-\cos\frac{2\pi\mu}{N}}.$$

Exercise 7.17. *Prove that for a square matrix M, $M^\dagger M = \mathbb{1}$ implies $MM^\dagger = \mathbb{1}$.*

$M^\dagger M = \mathbb{1} \qquad |\det M|^2 = 1 \qquad M^{-1} = M^\dagger \Rightarrow MM^\dagger = MM^{-1} = \mathbb{1}$

Exercise 7.19. *In Exercise 7.8, you should have found that $A_3 \cong \mathbb{Z}_3$ is a subgroup of S_3. You certainly know the irreducible representations of A_3 already; however, if I tell you that the character table of A_3 is hidden in the character table 7.1 of S_3, can you explain how you may extract the character table of A_3 from Table 7.1? Would the characters of $\bar{1}$ contribute to the character table of A_3? Hint: An irreducible representation of a group may be reducible as a representation of a subgroup.*

	1	$\bar{1}$	2
$[e]$	1	1	2
$[(12)]$	1	-1	0
$[(123)]$	1	1	-1

A_3是Abelian群，只有1d irrep. 每个元素自成一类，

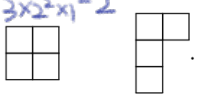
必有trivial rep：1.

假设$\bar{1}$也是A_3的一个irrep. 利用1与$\bar{1}$正交：

$1\times(1\times1)+1\times(1\times-1)+1\times(1\times1)\neq0$

因此$\bar{1}$ does not contribute to the character table of A_3.

Exercise 7.21. *Consider the following two Young tableaux of S_4 and find the basis vectors of the corresponding irreducible representations.*

$d=\frac{4!}{3\times2^2\times1}=2$  $d=\frac{4!}{4\times3\times2\times1}=1$

$\begin{array}{|c|c|}\hline 1&3\\\hline 2&4\\\hline\end{array}$ 轮换结构 $(12)(34)$，关于行对称，关于列反称.

$\rightarrow\ \downarrow$ $|1324\rangle+|3124\rangle+|1342\rangle+|3142\rangle$

$-(|2314\rangle+|1423\rangle+|3214\rangle+|4123\rangle+|2341\rangle+|1432\rangle$

$+|3241\rangle+|4132\rangle)+|2413\rangle+|4213\rangle+|2431\rangle+|4231\rangle$

$\begin{array}{|c|c|}\hline 4&3\\\hline 2&1\\\hline\end{array}$ $\longleftrightarrow$ 上式交换14

$\begin{array}{|c|c|}\hline 1&2\\\hline 3&\\\hline 4&\\\hline\end{array}$ $\longleftrightarrow$ $|1234\rangle+|2134\rangle-\sum_{P(134)}(-1)^P|P(1)2P(3)P(4)\rangle$

$-\sum_{P(134)}(-1)^P|2P(1)P(3)P(4)\rangle$

Exercise 7.23. *Prove that given $g, g' \in G$, if $g' = h^{-1}gh$ for some $h \in G$, then $g'^2 = h'^{-1}g^2h'$ for some $h' \in G$.*

$$g'^2 = h^{-1}ghh^{-1}gh = h^{-1}g^2h \qquad h' = h$$

Exercise 7.25. *Study the action of the projector (7.58) on a function $F : G \to \mathbb{C}$. You may expand the function in terms of its components in the irreducible representations of G.*

$$(7.58) \quad P_\mu = \frac{d_\mu}{|G|}\sum_{g\in G}\chi^\mu(g)^*\rho(g) = \frac{d_\mu}{|G|}\sum_{g\in G}\langle\chi^\mu|g\rangle\langle g|\rho\rangle$$

$$F(g) = \langle g|F\rangle = \sum_\mu \langle g|\chi^\mu\rangle\langle\chi^\mu|F\rangle$$

$$= \sum_\mu \chi^\mu(g)$$

$$P_\mu F(g) = \frac{d_\mu}{|G|}\sum_{f\in G}\sum_\nu \langle\chi^\mu|f\rangle\langle f|\rho\rangle\langle g|\chi^\nu\rangle\langle\chi^\nu|F\rangle$$

$$= \frac{d_\mu}{|G|}\sum_{f\in G}\sum_\nu \langle\chi^\mu|\chi^\nu\rangle\langle\chi^\nu|\rho\rangle\langle g|f\rangle\langle f|F\rangle$$

$$\sum_{g\in G}\sqrt{\frac{d_\mu}{|G|}}\rho^\mu(g)^*_{ab}\sqrt{\frac{d_\nu}{|G|}}\rho^\nu(g)_{kl} = \delta^{\mu\nu}\delta_{ak}\delta_{bl}.$$

$$= \frac{d_\mu}{|G|}\sum_{f\in G}\sum_\nu \delta^{\mu\nu} d_\nu\, \chi^\nu(\rho(g))\,\delta_{gf}\,F(f)$$

$$= \frac{d_\mu}{|G|} d_\mu \chi^\mu(\rho(g))F(g) \qquad \mu \text{ not for 求和}$$

$$P_\mu = \frac{d_\mu}{|G|}\sum_{g\in G}\chi^\mu(g)^*\rho(g) = \frac{d_\mu}{|G|}\sum_{g\in G}\rho^\mu(g)^*_{aa}\rho(g)$$

$$F(g) = \langle F|g\rangle = \langle F|\rho^R(g)|e\rangle = \sum_{\mu=1}^{nr}\sum_{a,b=1}^{d_\mu} F^\mu_{ab}\rho^\mu(g)_{ab}$$

$$P_\mu F(g) = \frac{d_\mu}{|G|}\sum_{g\in G}\sum_{\nu=1}^{nr}\sum_{a,b=1}^{d_\nu}\rho^\mu(g)^*_{cc}\rho(g)F^\nu_{ab}\rho^\nu(g)_{ab}$$

$$= \sqrt{\frac{d_\mu}{|G|}}\sum_{\nu=1}^{nr}\sum_{a,b=1}^{d_\nu}\sum_{g\in G}\left[\sqrt{\frac{d_\mu}{|G|}}\rho^\mu(g)^*_{cc}\rho^\nu(g)_{ab}\sqrt{\frac{d_\nu}{|G|}}\right]\sqrt{\frac{|G|}{d_\nu}}F^\nu_{ab}\rho(g)$$

$$= \sqrt{\frac{d_\mu}{|G|}}\sum_{\nu=1}^{nr}\sum_{a,b=1}^{d_\nu}\delta^{\mu\nu}\delta_{ca}\delta_{cb}\sqrt{\frac{|G|}{d_\nu}}F^\nu_{ab}\rho(g)$$

$$= \sum_{c=1}^{d_\mu} F^\mu_{cc}\rho(g) \qquad (7.26)$$

Exercise 7.27. *We can construct a finer projector*

$$P_{\mu,jk} = \frac{d_\mu}{|G|} \sum_{g\in G} \rho^\mu(g)^*_{jk} \rho(g). \tag{7.59}$$

Prove the idempotency:

$$P_{\mu,jk} P_{\nu,mn} = P_{\mu,km} \delta_{\mu\nu} \delta_{jn}. \quad \sum_{g\in G} \sqrt{\frac{d_\mu}{|G|}} \rho^\mu(g)^*_{ab} \sqrt{\frac{d_\nu}{|G|}} \rho^\nu(g)_{kl} = \delta^{\mu\nu}\delta_{ak}\delta_{bl}.$$

Give a state $|w\rangle$, what does this projector do?

$$P_{\mu,jk} P_{\nu,mn} = \frac{d_\mu}{|G|} \frac{d_\nu}{|G|} \sum_{g,f\in G} \rho^\mu(g)^*_{jk} \rho(g) \rho^\nu(f)^*_{mn} \rho(f)$$

$$= \sqrt{\frac{d_\mu}{|G|}} \sqrt{\frac{d_\nu}{|G|}} \sum_{g\in G} \sum_{f\in G} \sqrt{\frac{d_\mu}{|G|}} \rho^\mu(g)^*_{jk} \sqrt{\frac{d_\nu}{|G|}} \rho^\nu(f)^*_{mn} \rho(gf)$$

$$\overset{f\to g^{-1}}{=} \sqrt{\frac{d_\mu}{|G|}} \sqrt{\frac{d_\nu}{|G|}} \sum_{g\in G} \sum_{g^{-1}\in G} \sqrt{\frac{d_\mu}{|G|}} \rho^\mu(g)^*_{jk} \sqrt{\frac{d_\nu}{|G|}} \rho^\nu(g^{-1})^*_{mn} \rho(e)$$

$$\underset{\text{rep}}{\overset{\text{Unitary}}{=}} \sqrt{\frac{d_\mu}{|G|}} \sqrt{\frac{d_\nu}{|G|}} \sum_{g\in G} \sum_{g^{-1}\in G} \sqrt{\frac{d_\mu}{|G|}} \rho^\mu(g)^*_{jk} \sqrt{\frac{d_\nu}{|G|}} \rho^\nu((g^{-1})^{-1})_{nm} \rho(e)$$

$\rho^\nu(f)^*_{mn} = [\rho^\nu(f)^\dagger]_{nm}$

酉表示 $\rho^\nu(f)^\dagger = \rho^\nu(f)^{-1}$

$= \rho^\nu(f^{-1})$

$$= \sqrt{\frac{d_\mu}{|G|}} \sqrt{\frac{d_\nu}{|G|}} \sum_{g^{-1}\in G} \delta^{\mu\nu} \delta_{jn} \delta_{km} \rho(e)$$

$$P_{\mu,km} \delta_{\mu\nu} \delta_{jn} = \frac{d_\mu}{|G|} \sum_{g\in G} \rho^\mu(g)^*_{km} \rho(g) \delta_{\mu\nu} \delta_{jn}$$

Exercise 7.29. *Using the projector P above, find the two linearly independent motion modes. Draw the pictures. Be aware of the time-reversal invariance. Note that these two modes are the 2-fold degenerate eigenstates of H with the same degenerate eigen-frequency.*

$$P = P_2 - P_{T_x} - P_{T_y} = \begin{pmatrix} \frac{1}{3} & 0 & -\frac{1}{6} & -\frac{\sqrt{3}}{6} & -\frac{1}{6} & \frac{\sqrt{3}}{6} \\ 0 & \frac{1}{3} & \frac{\sqrt{3}}{6} & -\frac{1}{6} & -\frac{\sqrt{3}}{6} & -\frac{1}{6} \\ -\frac{1}{6} & \frac{\sqrt{3}}{6} & \frac{1}{3} & 0 & -\frac{1}{6} & -\frac{\sqrt{3}}{6} \\ -\frac{\sqrt{3}}{6} & -\frac{1}{6} & 0 & \frac{1}{3} & \frac{\sqrt{3}}{6} & -\frac{1}{6} \\ -\frac{1}{6} & -\frac{\sqrt{3}}{6} & -\frac{1}{6} & \frac{\sqrt{3}}{6} & \frac{1}{3} & 0 \\ \frac{\sqrt{3}}{6} & -\frac{1}{6} & -\frac{\sqrt{3}}{6} & -\frac{1}{6} & 0 & \frac{1}{3} \end{pmatrix}$$

eigenvectors

$$\begin{pmatrix} \frac{\sqrt{3}}{2} & -\frac{1}{2} & -\frac{\sqrt{3}}{2} & -\frac{1}{2} & 0 & 1 \\ -\frac{1}{2} & -\frac{\sqrt{3}}{2} & -\frac{1}{2} & \frac{\sqrt{3}}{2} & 1 & 0 \\ -\frac{\sqrt{3}}{2} & \frac{1}{2} & 0 & 0 & 0 & 1 \\ \frac{1}{2} & \frac{\sqrt{3}}{2} & 0 & 0 & 1 & 0 \\ \frac{\sqrt{3}}{2} & \frac{1}{2} & 0 & 1 & 0 & 0 \\ \frac{1}{2} & -\frac{\sqrt{3}}{2} & 1 & 0 & 0 & 0 \end{pmatrix}$$

eigen value $=1$

mode 1: $(\frac{\sqrt{3}}{2}, -\frac{1}{2})$

$(-\frac{\sqrt{3}}{2}, \frac{1}{2})$

$(\frac{\sqrt{3}}{2}, \frac{1}{2})$

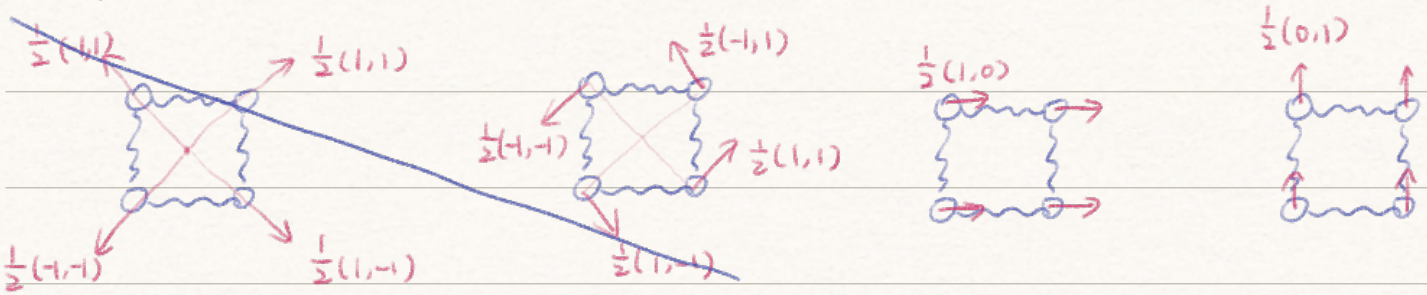
mode 2: $(-\frac{1}{2}, -\frac{\sqrt{3}}{2})$

$(\frac{1}{2}, \frac{\sqrt{3}}{2})$

$(\frac{1}{2}, -\frac{\sqrt{3}}{2})$

Exercise 7.31. *Use group-theoretical method to analyze the harmonic system in Fig. 7.7 and find the zero modes.*

S_4: $d = 1, 1, 2, 3, 3$ $8 = 2 \otimes (3+1)$



S_4作用下只有2个平凡表示？

Exercise 7.33. *Verify that the C-G coefficients we have obtained for decomposing the tensor product representation $2 \otimes 2$ of S_3 indeed form a 4×4 unitary matrix.*

$C^{\alpha J m}_{j_1 j_2 m_1 m_2} = \langle j_1, m_1; j_2, m_2 | J, m \rangle_\alpha$ $\quad (C_{j_1 j_2})^{\alpha J m}_{m_1 m_2}$ is unitary

即 $\sum_\alpha \sum_J \sum_m C^{\alpha J m}_{j_1 j_2 m_1 m_2} C^{j_1 j_2 m_1' m_2'}_{\alpha J m} = \delta^{m_1'}_{m_1} \delta^{m_2'}_{m_2}$

$\sum_{m_1} \sum_{m_2} C^{j_1 j_2 m_1 m_2}_{\alpha J m} C^{\alpha' J' m'}_{j_1 j_2 m_1 m_2} = \delta^{\alpha'}_{\alpha} \delta^{J'}_{J} \delta^{m'}_{m}$

$|j_1, m_1; j_2, m_2\rangle = \sum_\alpha \sum_J \sum_m C^{j_1 j_2 m_1 m_2}_{\alpha J m} |J, m\rangle_\alpha$

$|1,1\rangle = \pm \frac{1}{\sqrt{2}} (|2,1;2,1\rangle + |2,2;2,2\rangle)$

$|\bar{1},1\rangle = \pm \frac{1}{\sqrt{2}} (|2,1;2,2\rangle - |2,2;2,1\rangle)$ $\quad |2222\rangle = \frac{1}{\sqrt{2}} (11 + 22)$

$|2,1\rangle = \mp \frac{1}{\sqrt{2}} (|2,1;2,2\rangle + |2,2;2,1\rangle)$ $\quad |2121\rangle = \frac{1}{\sqrt{2}} (11 - 22)$

$|2,2\rangle = \pm \frac{1}{\sqrt{2}} (|2,2;2,2\rangle - |2,1;2,1\rangle)$ $\quad |2122\rangle = \frac{1}{\sqrt{2}} (\bar{1}1 - 21)$

$C^{2211}_{11} = \frac{1}{\sqrt{2}}$ $\quad C^{2211}_{22} = -\frac{1}{\sqrt{2}}$ $\quad |2221\rangle = \frac{-1}{\sqrt{2}} (\bar{1}1 + 21)$

$C^{2212}_{\bar{1}1} = \frac{1}{\sqrt{2}} = -C^{2212}_{21}$ $\quad C^{2221}_{\bar{1}1} = -\frac{1}{\sqrt{2}} = C^{2221}_{21}$

$C^{2222}_{21} = 0$ $\quad C^{2211}_{22} = -\frac{1}{\sqrt{2}}$

$C^{2212}_{22} = 0$ $\quad C^{2221}_{22} = 0$

$$\begin{cases} C^{11}_{2211} = C^{11}_{2222} = \frac{1}{\sqrt{2}} \qquad C^{\bar{1}1}_{2212} = -C^{\bar{1}1}_{2221} = \frac{1}{\sqrt{2}} \\ C^{22}_{2222} = -C^{21}_{2212} = -C^{21}_{2221} = -C^{22}_{2211} = \frac{1}{\sqrt{2}} \\ C^{21}_{2222} = C^{22}_{2212} = C^{22}_{2221} = 0 \end{cases}$$

可验证 $\sum_J \sum_m C^{\alpha J m}_{j_1 j_2 m_1 m_2} C^{j_1 j_2 m_1' m_2'}_{\alpha J m} = \delta^{m_1'}_{m_1} \delta^{m_2'}_{m_2}$

$\sum_{m_1} \sum_{m_2} C^{j_1 j_2 m_1 m_2}_{\alpha J m} C^{\alpha' J' m'}_{j_1 j_2 m_1 m_2} = \delta^{\alpha'}_{\alpha} \delta^{J'}_{J} \delta^{m'}_{m}$

如 $\sum_{JM} C^{JM}_{2211} C^{2211}_{JM} = C^{11}_{2211} C^{2211}_{11} + C^{22}_{2211} C^{2211}_{22} = \frac{1}{\sqrt{2}} \times \frac{1}{\sqrt{2}} + (-\frac{1}{\sqrt{2}}) \times (-\frac{1}{\sqrt{2}}) = 1$

$\sum_{JM} C^{JM}_{2211} C^{2212}_{JM} = C^{11}_{2211} C^{2221}_{11} + C^{22}_{2211} C^{2221}_{22} = \frac{1}{\sqrt{2}} \times 0 + (-\frac{1}{\sqrt{2}}) \times 0 = 0$